MASHUK ELAHI

Marine Electrical Engineering · Engineering Digitalisation and AI-Assisted Design

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RESEARCH INTEREST

Verifiable AI-assisted generation of class-compliant marine electrical design deliverables.

I have spent sixteen years designing, energising and handing over electrical power systems on vessels and offshore assets. For the past year I have been building something alongside that work, on a live 42 m fast crew boat newbuild: a toolchain in which an AI agent reads the governing drawings and class rules and proposes engineering values, deterministic Python generators render every deliverable, and independent machine verifiers check the result before anything is issued. The engineer still decides. The machine does the reading, the rendering and the checking.

This is not a new direction for me. My master's capstone was already a work-based study of digitalising electrical commissioning at the same yard — telemetry, a digital twin and inventory traceability. The research below carries that from the commissioning floor back up into the design deliverables themselves.

It works. A 1,300-row cable schedule, a 40-drawing controlled set, an alarm point list and a drawing register now stay consistent with each other across revisions, because every value carries a record of where it came from and no build is issued until two independent builds agree byte for byte and the verifier passes. What I want to study is the part I had to invent as I went: where exactly the trust boundary belongs between an agent that reads and an engineer who signs, how the provenance of a single engineering value is recorded so that a machine can check it, and whether a pipeline built this way can be made acceptable to a classification society rather than merely useful to a yard.

THE RESEARCH I WANT TO DO

Working title: Provenance-checked generation of class-compliant marine electrical design deliverables.

The concept note accompanying this CV sets out the problem, the four research questions, a work programme and the demonstrator. In short: formalise the provenance model, build an open demonstrator that takes a vessel load list through to a verified cable schedule and alarm list, and evaluate it against a real newbuild set and against class and flag requirements.

ENGINEERING RESEARCH AND DEVELOPMENT — CURRENT WORK

Strategic Marine Pte. Ltd., Singapore | Electrical Engineer, Hybrid Marine & Vessel Electrification | Jul 2024 – Present

Hull H675, 42 m fast crew boat, Australian flag, Lloyd's Register SSC and AMSA NSCV. Alongside the design work I built and operate the digitalisation toolchain described above.

- Deterministic generation. Every deliverable — cable schedule, cable order quantity summary, electrical load list, AMS alarm point list, drawing register — is produced by a Python generator from a single data set rather than edited by hand. A release is accepted only when two independent builds are byte-identical.
- Machine gating. A manufacturability and compliance linter runs before every issue and must report zero errors. Its gates encode real rules: a 240 V circuit may not sit on a 150/250 V cable family; a 0.6/1 kV cable may not fall below 1.5 mm²; power multicores at and above 10 mm² take the flexible Class 5 construction.
- Provenance on every value. Each engineering figure carries its status — read from a named document and page, derived by a shown calculation, estimated, or not held — and a figure without a source or a derivation cannot reach a deliverable. A class clause is citable only when its page has been read.

- Independent verification. Releases are checked by read-only agents that did not produce the work. One full-set audit across roughly 650 drawing pages returned 327 findings, every one dispositioned before issue; defect families, not individual defects, became new standing checks.
- H675 Draw — a drawing program for the yard. A desktop application in Python and Qt with its own JSON drawing format, a symbol library of 610 symbols (419 cut from the project's own 31 drawings, 191 redrawn from the NFPA icon set), SVG / DXF / PDF rendering, a verifier of 15 checks, and 40 tools exposed so an AI agent can author and mark up sheets directly. All 40 project drawings render and verify; the verifier rejects deliberately tampered input.
- Geometry extraction. Cable tray layouts, bulkhead and deck penetrations and frame lines are read out of the arrangement DXF into a geometry model, and cable routes are derived from it rather than assigned by hand.
- Design work proper over the same period: LV distribution for a three-generator 450 ekW prime-power system, first-pass approval from four independent certification bodies with zero rework; integration of a 72 kWh lithium-ion BESS with its battery management system and DC/DC inverters, cutting fuel draw by up to 50% at low load.

EARLIER ENGINEERING EXPERIENCE

Exyte, Singapore | Senior Project Engineer, Critical-Power Systems & Prefabrication | Feb 2023 – Jul 2024

- 300 mm semiconductor cleanroom critical-power hook-up, energisation and functional performance testing on 30 MW (33 MVA) of critical load at 1.5 kW/m² fab power density; zero audit rework and first-pass PLC/SCADA interlock validation.

Phoenix Mecano S.E.A. Pte. Ltd., Singapore | E&I Engineer, Renewable, Cogeneration & Hazardous Areas | Jan 2021 – Jan 2023

- Malaysia's first solar-wind autonomous remote-power installation on a 24 V DC solar-battery bus, removing routine diesel-generator use and enabling unmanned operation — the same hybrid source-transfer problem as my undergraduate project, at plant scale.
- E&I scope for an onshore 10 MW cogeneration plant: transformers, MV/LV switchgear, a 400 kVA UPS, SCADA integration and instrument loops; IECEx/ATEX hazardous-area design.

Keppel FELS / Keppel Offshore & Marine, Singapore | E&I Superintendent, Offshore Power Systems & Commissioning | Nov 2012 – Dec 2020

- Planned and monitored the electrical cabling programme for a jack-up rig newbuild — a scope estimated at 430,000 m of cable and 365,500 manhours — tracking cabling metres and connection points by zone and phase against a 0.85 manhour-per-metre productivity target.
- Medium-voltage thruster power distribution on semi-submersible newbuilds: thruster transformers feeding MV7000 converters across two thruster rooms.
- E&I drawing and as-built package for an FPSO topsides scope: 11 kV switchgear connection diagrams, 440 V essential and process switchgear, fire-and-gas and PAGA diagrams across four modules.
- Energised an offshore-wind substation to full IEC 61850 protection-relay conformity with zero punch-list items; led MV 11 kV and ABB 800xA commissioning with crews of up to 100.
- Electrical schematics for an autonomous tug — ABB/Niigata propulsion interface, vessel-care system, 240 V autonomous power panel — and the electrical scope of a 12 MW, 3,750 m³ floating living-lab vessel.

Ananda Shipyard & Slipways Ltd., Bangladesh | Assistant Electrical Engineer | Sep 2009 – Oct 2012

- Electrical installation, testing and commissioning on vessel newbuilds.

ACADEMIC WORK

Master's dissertation — UCAM Universidad Católica de Murcia, Spain

“Integration of Short and Long Duration Energy Storage Systems with Renewable Energy and EV Charging across European Countries” — sole-authored systematic review, 40 pp; tutor Nieves Peña Bo. Storage sizing and duration matching against renewable generation profiles and charging demand, country by country.

MSc Digital Transformation (Pass with Merit) — London Metropolitan University, UK

Capstone: Work Based Project (MN7P99NU), Level 7, 40 credits / 20 ECTS, mark 68: “A Digital Transformation Strategy for Integrated Commissioning, Telemetry, and Inventory Control at Strategic Marine Pte. Ltd.”, supervised by Kushan Yapa. A work-based study of electrical commissioning at my own yard: telemetry capture during commissioning, a digital twin for virtual commissioning, and SKU-level inventory traceability through an integrated warehouse management system, delivered as a phased single-vessel pilot against defined KPIs and assessed with PESTEL, Porter’s Five Forces, McKinsey 7S, Kotter’s 8-step, Bridges’ transition model, Mendelow’s matrix and DICE.

MBA — University of Sunderland, UK

BSc final-year project — Bangladesh University, 2008

Design and construction of an automatic transfer switch with a backup control system for a hybrid power plant: automatic transfer of load between a primary supply and standby sources, with a backup controller holding critical load through the transfer. Built and tested as hardware. The project report is not retained. The same problem — holding supply continuity across an automatic source transfer — is what I now work on at vessel scale with generators, batteries and hybrid drivelines.

EDUCATION

- MSc Digital Transformation (Pass with Merit) — London Metropolitan University, UK — Apr 2026. FHEQ Level 7; capstone Work Based Project, 40 credits / 20 ECTS.
- Master of Business Administration (MBA) — University of Sunderland, UK — Jun 2025
- BSc Electrical and Electronic Engineering — Bangladesh University — Jul 2008
- Diploma in Electronics Engineering — Jessore Polytechnic Institute — Jan 2004
- Continuing-education award: Sustainable Electricity Generation (UCAM Universidad Católica de Murcia, Spain, 60 ECTS, título propio) — Apr 2026. A university-own award, listed separately from the degrees above.

STANDARDS AND CLASS FRAMEWORKS WORKED TO

Lloyd's Register Rules for Special Service Craft; AMSA NSCV Part C Section 5B; AS/NZS 3004.2, AS/NZS 3000, AS/NZS 3008; IEC 60092, IEC 61892, IEC 60364, IEC 61850, IEC 60332; ABS, Bureau Veritas and DNV class approval. IACS UR E26 / E27 vessel cyber-resilience and IEC 62443 are areas of active study rather than project-verified experience.

TECHNICAL SKILLS

Python (deterministic document and drawing generation, PDF content-stream editing, DXF geometry extraction with ezdxf, PyMuPDF, openpyxl); Qt/PySide6 desktop applications; AI agent tooling and tool-server interfaces; AutoCAD Electrical scripting; ABB 800xA; EPMS/SCADA and PLC/SCADA; IEC 61850 protection relays; FAT/SAT, FMEA, energisation and commissioning.

CERTIFICATIONS

- Certified Data Centre Professional (CDCP), EPI — 2026
- ISO 50001:2018 Energy Management Lead Auditor, GAQM / IGC
- ISO 9001:2015 Quality Management Lead Auditor, GAQM / IGC
- IECEx Ex 001: Basic Principles of Protection in Explosive Atmospheres — valid through Oct 2027

Referees: available on request.